

LAB EXPERIMENT RECORD

Set 22: Time Series Analysis

Department	B.Tech AIDS
Course Code / Course Name	DSA06 / Data Handling and Visualization
Experiment	Data Visualization using R
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Register Number	192524110
Date	16.10.2026
Day	Thursday

Aim

To analyze monthly sales as a time series in R using a time series line chart, a scatter plot, and an autocorrelation plot, and to build an interactive dashboard with a sales trend line chart and a product sales pie chart using R Shiny.

Dataset: Monthly Sales Data

Month	Sales (in \$)
January	15000
February	18000
March	22000
April	20000
May	23000

Note: The dataset does not include an advertising budget or product-wise sales. The programs use the same values as Set 1: Advertising Budget (\$) – 2000, 2500, 3200, 2800, 3500 (January to May); Yearly Product Sales (\$) – Product A 120000, Product B 95000, Product C 80000, Product D 65000, Product E 50000. The months are taken as January to May 2025 for the time series.

PROGRAM 1: Time Series Line Chart of Monthly Sales

In R, create a time series line chart to visualize the trend in monthly sales. Label the axes and title the chart.

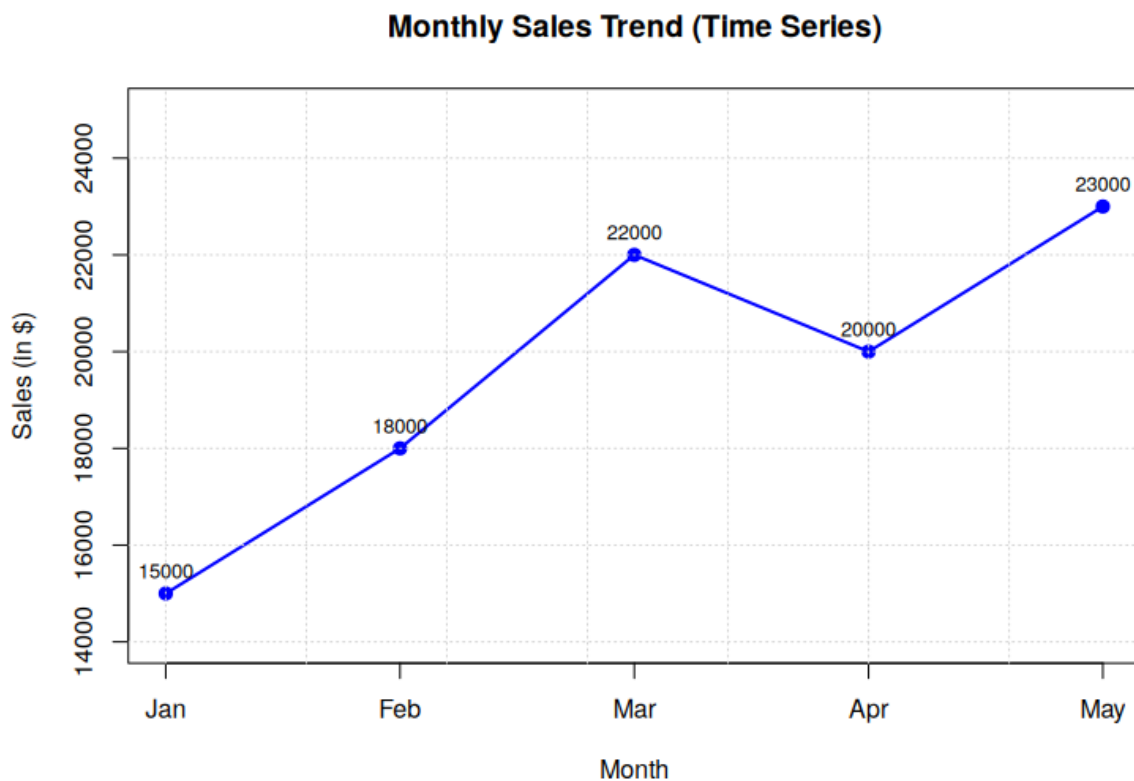
```
# Create dataset for Set 22
monthly_sales <- data.frame(
  Month = c("January", "February", "March", "April", "May"),
  Sales = c(15000, 18000, 22000, 20000, 23000),
  Ad_Budget = c(2000, 2500, 3200, 2800, 3500) # Advertising budget for Q2
)

print(monthly_sales)

# Convert sales into a monthly time series (January 2025 to May 2025)
sales_ts <- ts(monthly_sales$Sales, start = c(2025, 1), frequency = 12)
print(sales_ts)
```

1. Time Series Line Chart

```
plot(sales_ts,
     type = "o",
     col = "blue",
     lwd = 2,
     pch = 19,
     xaxt = "n",
     ylim = c(14000, 25000),
     xlab = "Month",
     ylab = "Sales (in $)",
     main = "Monthly Sales Trend (Time Series)")
axis(1, at = time(sales_ts), labels = month.abb[1:5])
text(time(sales_ts), sales_ts, labels = sales_ts, pos = 3, cex = 0.8)
grid()
```



Output 1: Monthly Sales Trend (Time Series)

Observation: Sales rise from \$15,000 in January to \$22,000 in March, dip to \$20,000 in April, and reach the highest value of \$23,000 in May. Overall, the time series shows an upward trend, with a total increase of about 53% over five months.

PROGRAM 2: Scatter Plot of Advertising Budget vs Monthly Sales

In R, generate a scatter plot to analyze the relationship between advertising budget and monthly sales. Explain any insights.

```
# Create dataset for Set 22
monthly_sales <- data.frame(
  Month = c("January", "February", "March", "April", "May"),
  Sales = c(15000, 18000, 22000, 20000, 23000),
  Ad_Budget = c(2000, 2500, 3200, 2800, 3500) # Advertising budget for Q2
)

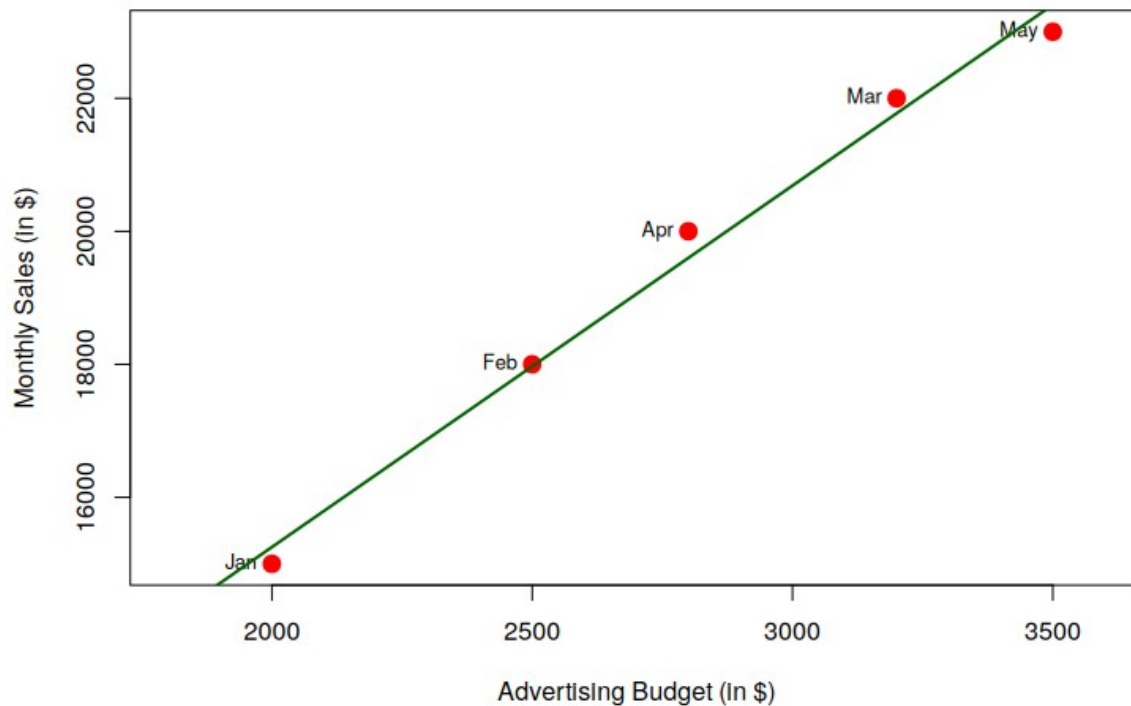
print(monthly_sales)

# 2. Scatter Plot of Advertising Budget vs Monthly Sales
plot(monthly_sales$Ad_Budget, monthly_sales$Sales,
     col = "red",
     pch = 19,
     cex = 1.5,
     xlim = c(1800, 3600),
     xlab = "Advertising Budget (in $)",
     ylab = "Monthly Sales (in $)",
     main = "Advertising Budget vs Monthly Sales")
text(monthly_sales$Ad_Budget, monthly_sales$Sales,
     labels = substr(monthly_sales$Month, 1, 3), pos = 2, cex = 0.8)

# Add a trend line
abline(lm(Sales ~ Ad_Budget, data = monthly_sales), col = "darkgreen", lwd = 2)

# Correlation
cor(monthly_sales$Ad_Budget, monthly_sales$Sales)
```

Advertising Budget vs Monthly Sales



Output 2: Advertising Budget vs Monthly Sales

Insight: The points lie almost exactly on the upward trend line, showing a very strong positive relationship between advertising budget and sales (correlation about 0.99). Every additional \$1 of advertising is associated with about \$5.43 more in monthly sales. The drop in April sales matches the lower April budget (\$2,800). With only five months of data, this shows association and not proof of cause.

PROGRAM 3: Interactive Dashboard for Sales Trend and Product Distribution

In Tableau, build a dashboard combining a line chart showing sales trend and a pie chart displaying the distribution of sales across products.

Note: Tableau could not be installed on the system used for this experiment, so the interactive dashboard was developed in R using the Shiny package, with the same charts and interactive filters required in the question.

```
library(shiny)

# Dataset for Set 22
monthly_sales <- data.frame(
  Month = c("January", "February", "March", "April", "May"),
  Sales = c(15000, 18000, 22000, 20000, 23000)
)
# Yearly product sales used for the pie chart
product_sales <- data.frame(
  Product = c("Product A", "Product B", "Product C", "Product D", "Product E"),
  Sales = c(120000, 95000, 80000, 65000, 50000)
)
prod_cols <- c("steelblue", "darkorange", "seagreen", "orchid", "gold")
names(prod_cols) <- product_sales$Product

ui <- fluidPage(
  titlePanel("Sales Trend and Product Distribution Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      checkboxGroupInput("months", "Months:", choices = monthly_sales$Month,
        selected = monthly_sales$Month),
      checkboxGroupInput("products", "Products:", choices = product_sales$Product,
        selected = product_sales$Product)
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("linePlot", height = "340px")),
        column(6, plotOutput("piePlot", height = "340px"))
      )
    )
  )
)

server <- function(input, output) {
  output$linePlot <- renderPlot({
    d <- monthly_sales[monthly_sales$Month %in% input$months, ]
    idx <- match(d$Month, monthly_sales$Month)
    plot(idx, d$Sales, type = "o", col = "blue", lwd = 2, pch = 19, xaxt = "n",
      xlim = c(0.7, 5.3), ylim = c(14000, 25000), xlab = "Month", ylab = "Sales ($)",
      main = "Monthly Sales Trend")
    axis(1, at = idx, labels = substr(d$Month, 1, 3))
    grid()
  }, width = 420, height = 340, res = 96)

  output$piePlot <- renderPlot({
    d <- product_sales[product_sales$Product %in% input$products, ]
    pct <- round(100 * d$Sales / sum(d$Sales), 1)
    par(mar = c(1, 1, 3, 1))
  })
}
```

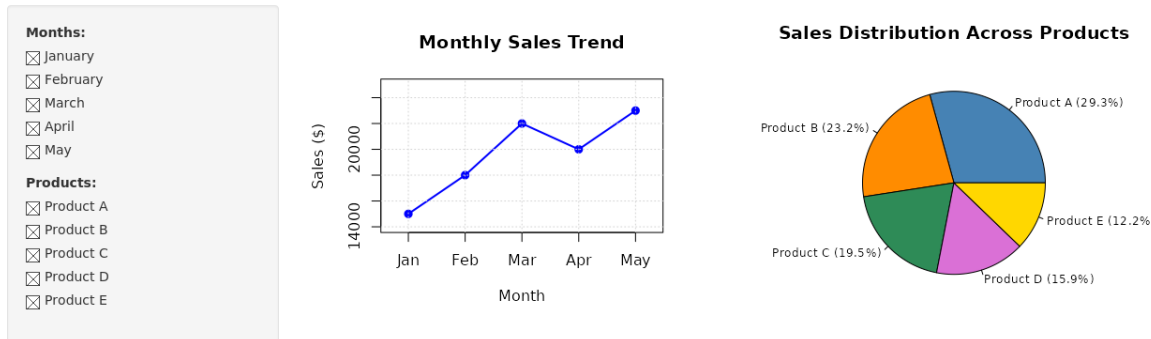
```

    pie(d$Sales, labels = paste0(d$Product, " (", pct, "%)"), col = prod_cols[d$Product],
        main = "Sales Distribution Across Products", cex = 0.8)
  }, width = 420, height = 340, res = 96)
}

shinyApp(ui = ui, server = server)

```

Sales Trend and Product Distribution Dashboard



Output 3: Sales Trend and Product Distribution Interactive Dashboard (R Shiny)

Observation: The dashboard shows the monthly sales trend and the product sales pie chart side by side. The Months filter chooses which months appear on the line chart, and the Products filter chooses which products appear on the pie chart, with the percentage shares recalculated for the selected products. With all products selected, Product A has the largest share (29.3%).

PROGRAM 4: Autocorrelation Plot of Monthly Sales

In R, create an autocorrelation plot to identify seasonality in the time series data.

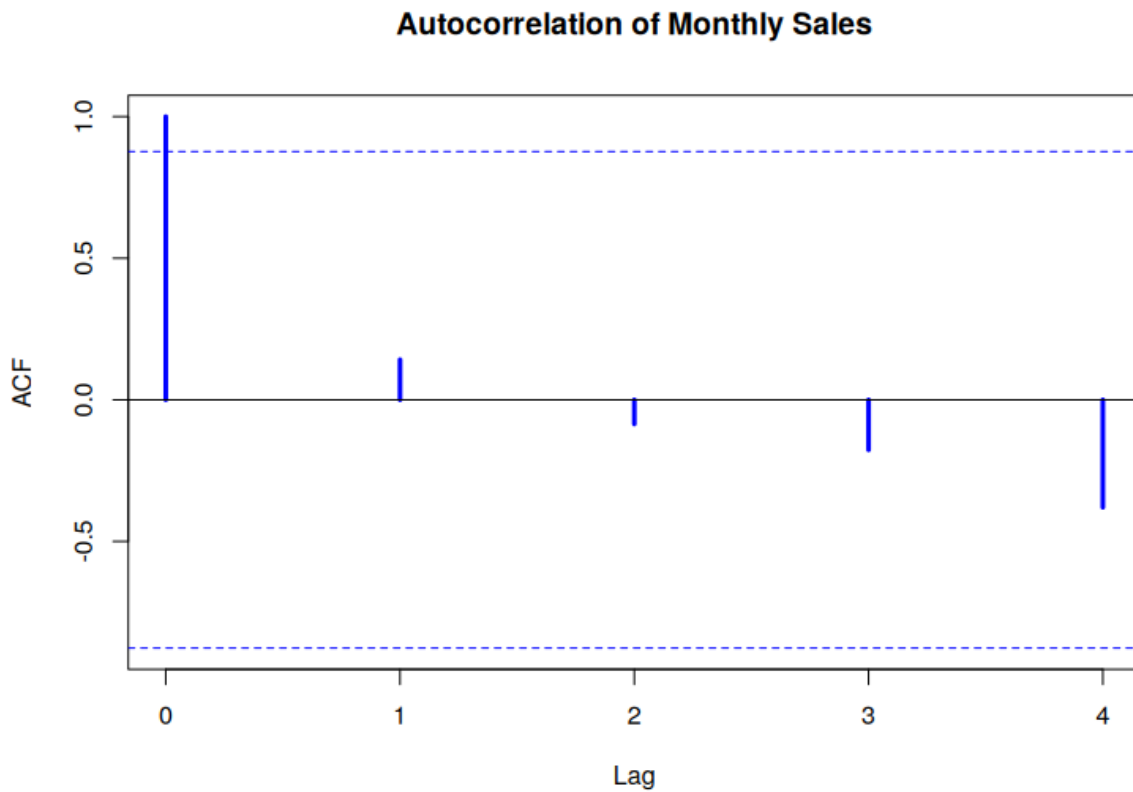
```
# Create dataset for Set 22
monthly_sales <- data.frame(
  Month = c("January", "February", "March", "April", "May"),
  Sales = c(15000, 18000, 22000, 20000, 23000),
  Ad_Budget = c(2000, 2500, 3200, 2800, 3500) # Advertising budget for Q2
)

print(monthly_sales)

# Convert sales into a monthly time series
sales_ts <- ts(monthly_sales$Sales, start = c(2025, 1), frequency = 12)

# Autocorrelation values
acf_values <- acf(sales_ts, lag.max = 4, plot = FALSE)
print(acf_values)

# 4. Autocorrelation Plot
acf(as.numeric(sales_ts),
    lag.max = 4,
    col = "blue",
    lwd = 3,
    main = "Autocorrelation of Monthly Sales")
```



Output 4: Autocorrelation of Monthly Sales

Observation: The autocorrelation is 1 at lag 0, as expected, and small at the other lags: 0.14 at lag 1, -0.09 at lag 2, -0.18 at lag 3, and -0.38 at lag 4. All of these lie inside the blue significance limits, so there is no significant autocorrelation and no seasonal pattern can be detected. Five months of data is too short to identify seasonality, which would need at least two full years of monthly data.

Procedure

1. Create the monthly sales data frame with Month, Sales, and Advertising Budget.
2. Convert the sales into a monthly time series using `ts()` and draw the time series line chart using `plot()`.
3. Use `plot()` for the advertising budget vs sales scatter plot, `abline()` with `lm()` for the trend line, and `cor()` for the correlation.
4. Install and load the shiny package, and create the dashboard layout using `fluidPage()`, sidebar filters, and output areas for the charts.
5. Write the server function to filter the data with `reactive()` and redraw the outputs, then run the dashboard using `shinyApp()` and test the filters.
6. Use `acf()` to calculate and plot the autocorrelation of the sales series.

Result

The monthly sales data was successfully analyzed in R using a time series line chart, a scatter plot, and an autocorrelation plot, and an interactive dashboard combining the sales trend line chart and the product sales pie chart was built using R Shiny.

Key R Functions Used

Function	Purpose
<code>ts()</code>	Converts the sales values into a monthly time series.
<code>plot()</code>	Creates the time series line chart and scatter plot.
<code>axis()</code> / <code>text()</code>	Adds month labels and value labels.
<code>abline()</code> / <code>lm()</code>	Adds a linear trend line.
<code>cor()</code>	Calculates the correlation coefficient.
<code>acf()</code>	Calculates and plots the autocorrelation.
<code>pie()</code>	Creates the product sales pie chart on the dashboard.
<code>fluidPage()</code> / <code>checkboxGroupInput()</code>	Creates the dashboard layout and filters.
<code>reactive()</code> / <code>renderPlot()</code>	Filters the data and redraws the charts.
<code>shinyApp()</code>	Runs the interactive dashboard.